Be sure you see that the derivative of a function of x is also a function of x. This "new" function gives the slope of the tangent line to the graph of f at the point $(x, f(x))$, provided the graph has a tangent line at this point.

The process of finding the derivative of a function is called **differentiation**. A function is **differentiable** at x if its derivative exists at x and is **differentiable on an open interval** (a, b) if it is differentiable at every point in (a, b).

In addition to $f'(x)$, which is read as 'f prime of x' or 'f dash of x', other notations are used to denote the derivative of $y = f(x)$. The most common notations are

$f'(x), \frac{dy}{dx}, y', \frac{d}{dx}[f(x)], D_x[y]$ or Dy or y_1. Here $\frac{d}{dx}$ or D is the differential operator.

The notation $\frac{dy}{dx}$ is read as "derivative of y with respect to x" or simply "dy – dx", or we should rather read it as "*Dee y Dee x*" or "*Dee Dee x* of *y*". But it is cautioned that we should not regard $\frac{dy}{dx}$ as the quotient $dy \div dx$ and should not refer it as "*dy by dx*". The symbol $\frac{dy}{dx}$ is known as **Leibnitz symbol.**

10.2.4 One sided derivatives (left hand and right hand derivatives)

For a function $y = f(x)$ defined in an open interval (a, b) containing the point x_0, the **left hand and right hand derivatives** of f at $x = x_0$ are respectively denoted by $f'(x_0^-)$ and $f'(x_0^+)$, are defined as

$$f'(x_0^-) = \lim_{\Delta x \to 0^-} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

and $$f'(x_0^+) = \lim_{\Delta x \to 0^+} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$, provided the limits exist.

That is, the function is differentiable from the left and right. As in the case of the existence of limits of a function at x_0, it follows that $f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ exists if and only if both $f'(x_0^-) = \lim_{\Delta x \to 0^-} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ and $f'(x_0^+) = \lim_{\Delta x \to 0^+} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ exist and $f'(x_0^-) = f'(x_0^+)$.

Therefore $f'(x_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$ if and only if $f'(x_0^-) = f'(x_0^+)$.

If any one of the condition fails then f is not differentiable at x_0.

In terms of $h = \Delta x > 0$,

$$f'(x_0^+) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$ and

$$f'(x_0^-) = \lim_{h \to 0} \frac{f(x_0 - h) - f(x_0)}{-h}$$.

10_11th-Maths-Vol2-Unit10.indd 143 21/11/2024 15:05:35